CIRCUIT THEORY II - R4053 - 2025	
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T.P. LAB N° 2 : Quadrupole Theory	
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1. INTRODUCTION

This laboratory work aims to apply basic concepts seen throughout the second semester on quadrupoles, Z-Y-T-IMG-S parameters; and through an appropriate design, simulate and implement a resistive attenuator and a constant impedance filter.

1.1 Goals

- Become familiar with resistive attenuator structures and constant impedance filters.
- Design adapted systems using concepts from classical theory.
- Use the concepts seen throughout the second semester of quadrupole theory, parameters Z, Y, T, Image and S, synthesis and analysis.

1.2 Part One

The first part deals with the design of a dissipative attenuator, using more than one stage, to reduce the amplitude of a signal while maintaining matching of impedances and a maximum attenuation error of$\pm\ 1db$considering the Standardization of components to commercial values. The group's conditions can be observed in Figure 1.

Grupo	Atenuación [dB]	Zi1 [Ω]	Zi2 [Ω]
1	20	50	50
2	22	50	50
3	24	50	50
5	18	50	50

Gráfico 1

Requested conditions.

The implementation should be on a board using terminals to connect the measuring instruments and the terminal load, facilitating measurements in the laboratory.

1.3 Part two

In a second stage, a bridged T-type constant impedance filter is developed that must comply with the following voltage transfer:

$$H(S) = \frac{V2}{V1} = \frac{S}{S^2+S+1}$$

It must be measured under load condition at both ports.50Ω, with a central frequency of$fc = 1100kHz$.

2. DISSIPATIVE ATTENUATOR

It is requested $Z0 = 50\Omega$ with an attenuation of -$24dB$.
It was decided to design a T-bridge based on the criteria of the chair.

2.1 Calculations

It is shown that the transfer of a T-bridged circuit has a transfer as follows: $\frac{V2}{V1} = \frac{R_0}{R_3 + R_0}$ where $R_0^2 = R_3 * R_4$ and with $R_1 = R_2 = R_G = R_L = R_0$

Developing:

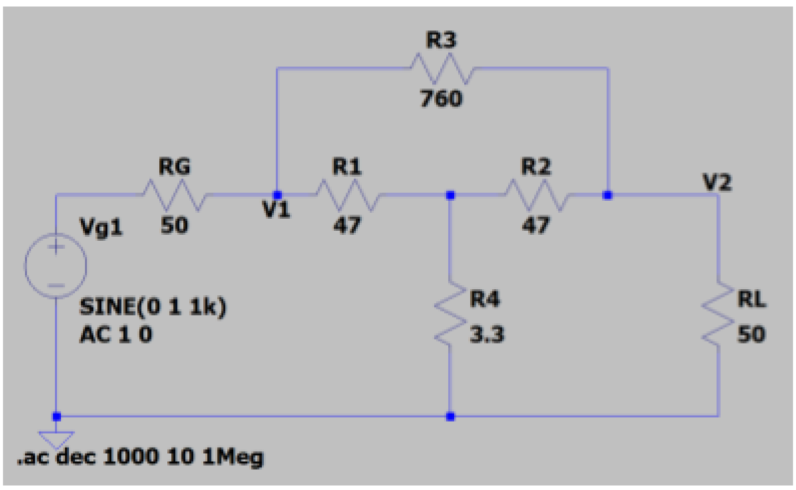
$$\frac{V2}{V1} = \frac{R_0}{R_3 + R_0} = -24dB = 20log(At[veces]) => At[veces] = 63,096 * 10^{-3}$$

$$R_3 = \frac{R_0 - At[veces]*R_0}{At[times]} = 742,45\Omega \quad \& \quad R_4 = \frac{R_0^2}{R_3} = 3.37\Omega$$

When converted to commercial values:

$$R_1 = R_2 = 47\Omega \quad \& \quad R_3 = 330\Omega + 330\Omega + 100\Omega \quad \& \quad R_4 = 3,3\Omega$$

The resulting circuit can be seen in Figure 2.



Circuit chosen for the dissipative attenuator.

Gráfico 2

2.2 Simulation

The DC behavior of the circuit is presented by simulating the values obtained previously in graph 3.

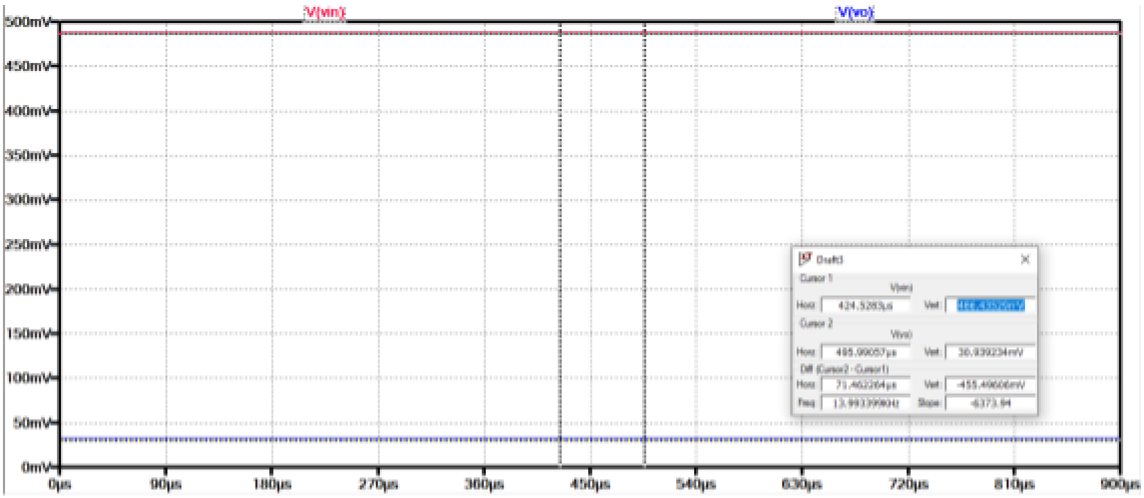


Gráfico 3

DC response with Vin = 486.44mV and Vo = 30.94mV.

From which it follows that:

$$\frac{Vo}{Vin} = \frac{30.94mV}{486.44mV} = 0.0636V/V \quad \rightarrow \quad 20 * log(\frac{Vo}{Vin}) = -23.93\text{dB}$$

Figure 4 shows the frequency response of the attenuator.

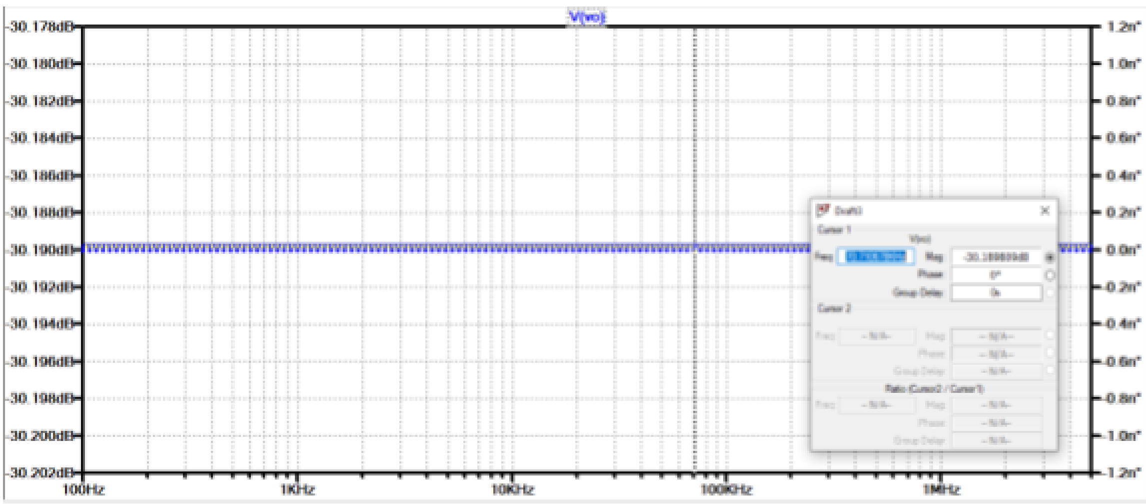


Gráfico 4

Attenuator attenuation.

2.3 S Parameters

Since the circuit is both passive and symmetrical, it follows that:

$$S_{11} = S_{22} \quad \& \quad S_{12} = S_{21} \quad \& \quad S_{11} = \frac{R_0 - Z_i}{Z_i + R_0} \quad \& \quad S_{21} = 2 * \frac{V_0}{V_g} * \sqrt{\frac{Z_{02}}{Z_{01}}}$$

where adaptation demonstrates that $Z_{01} = Z_{02} = R_0$

The input voltage and current values were obtained through simulation in graph 5.

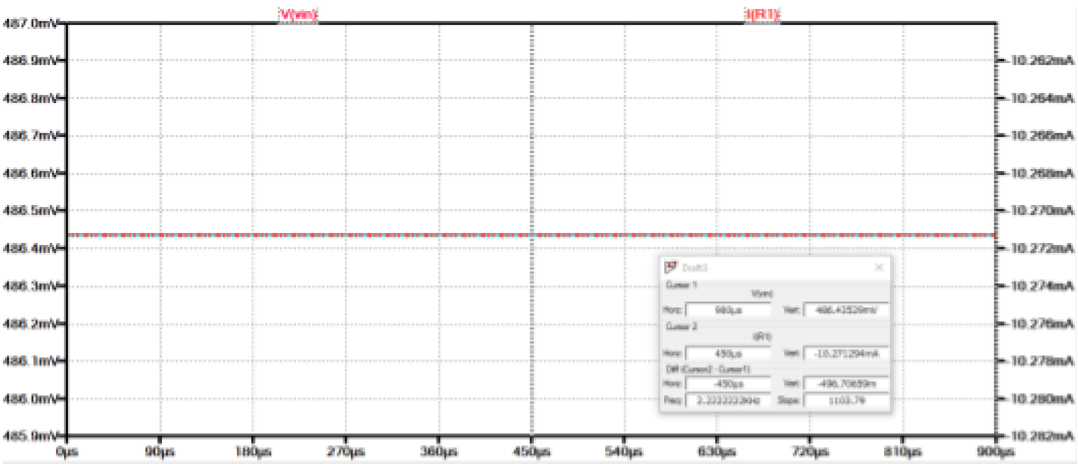


Gráfico 5

Tensión de entrada (Vin = 486.44mV) y corriente de entrada (Iin = 10.27mV).

Taking into account that $Z_i = \frac{V_i}{I_i} = \frac{486.44mV}{10.27mV} \simeq 47,36$

$$S_{11} = \frac{R_0 - Z_i}{Z_i + R_0} = \frac{50 - 47,365}{50 + 47,365} = \frac{2,635}{97,365} = 0,0271 = S_{22}$$

$$S_{21} = 2 * \frac{V_0}{V_g} * \sqrt{\frac{Z_{02}}{Z_{01}}} = 2 * \frac{0,03}{1} * \sqrt{\frac{50}{50}} = 0,06$$

2.4 Measurements

To verify the operation of the proposed circuit, various measurements were taken as requested by the chair.

Part A

Ten points were taken in the frequency range of the function generator to then validate the attenuator's modulus response when loaded with$Z0$.

Using a sinusoidal signal of$1Vpp$ From the generator, it was observed that the voltage obtained at the circuit input was clearly attenuated by the impedance of the instrument used. Table 1 presents the measured IN/OUT voltage values for a given frequency variation.

Frequency IN [Hz]	***IN Amplitude [mVpp]***	***Amplitud OUT [mVpp]***
50	468	32
100	444	32.7
250	480	31.6
500	468	32
750	480	30.3
900	472	32.7
1k	472	32.7
2k5	472	32.4
3k	472	30.3
5k	468	31.2

Tabla 1

On the other hand, in Table 2 these values were used to obtain the final attenuation in dB.

Frequency [Hz]	***Attenuation [dB]***
50	-23.3
100	-22.66
250	-23.63
500	-23.3
750	-23.99
900	-23.19
1k	-23.19
2k5	-23.27
3k	-23.85
5k	-23.52

Tabla 2

In graph 6 we present the comparison of the measurements made in the laboratory with what was expected in the simulation.

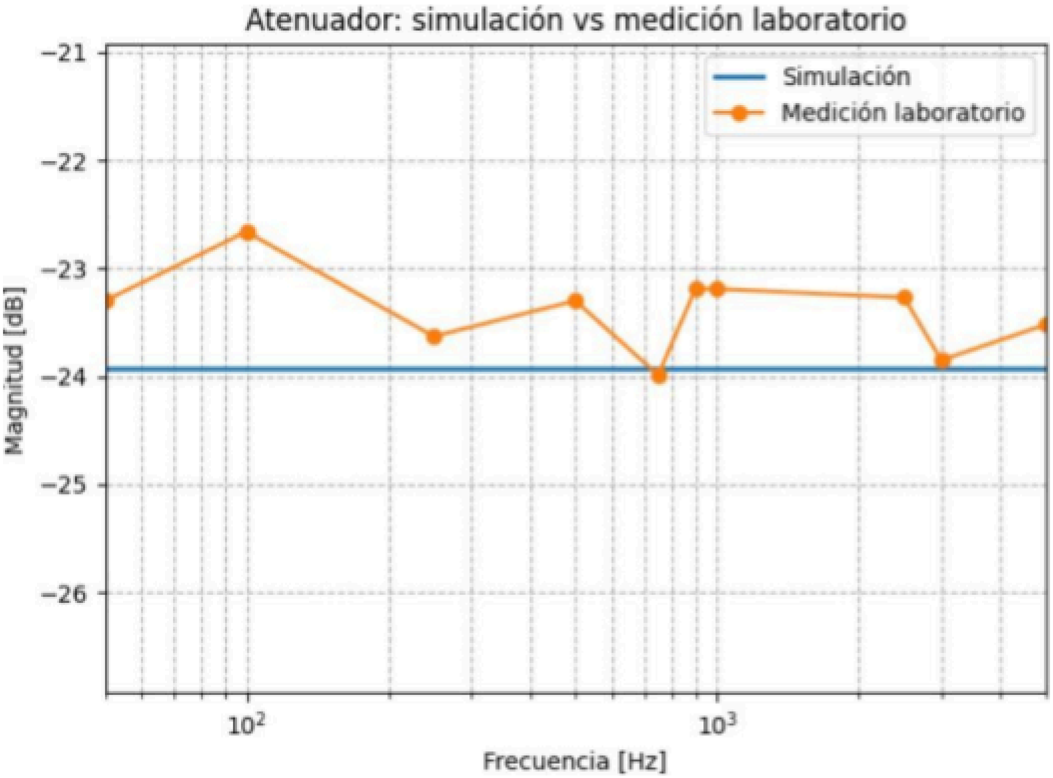


Gráfico 6

Comparison between simulated and measured values in the laboratory.

Attached below are images 1 and 2 of measurements taken from the oscilloscope.

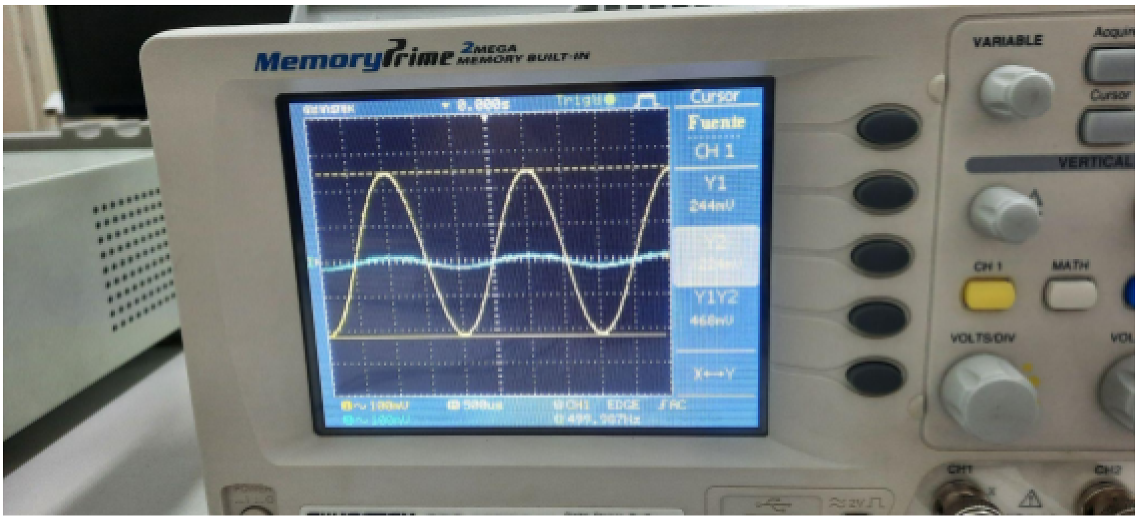


Imagen 1

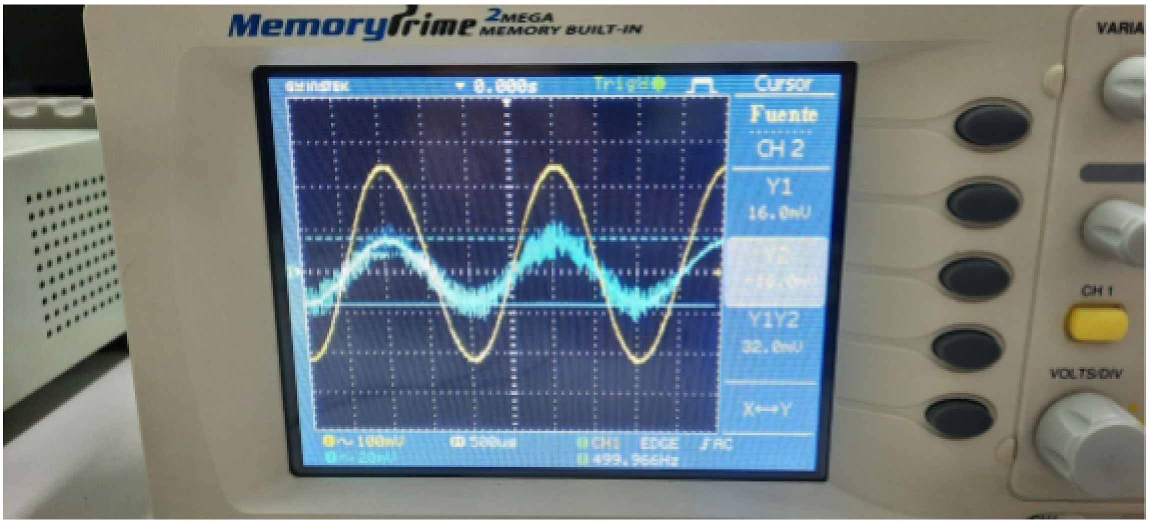


Imagen 2

Oscilloscope measurement of the voltage applied to the load at 500Hz.

3. CONSTANT IMPEDANCE FILTER

It was decided to design a T-bridge based on the criteria of the chair, with the main difference being the values of$Za$and Zb.

3.1 Calculations

It is requested$Z0 = 50\Omega$and a central frequency of$1100\ kHz$Starting from a generic bandpass seen in class, it is denormalized for these conditions.

- $\omega_n = 2 * \Pi * f = 2 * \Pi * 1100k = 6,912 \frac{Mrad}{seg}$
- $Z_n = 50\Omega$

This way you can calculate the values of each component and then get the closest commercial value.

- $L = \frac{Z_n}{\omega_n} = 7,234\mathrm{uF}$
- $C = \frac{1}{\omega_n * Z_n} = \frac{1}{6,912 \frac{Mrad}{seg} * 50\Omega} = 2,894nF \rightarrow C = 3.3nF$

In the case of inductance, the coils were constructed at home with enameled wire and their actual value was measured using an LCR (*graphs 7 and 8*).

Gráfico 7

Measurement of the first coil (L = 7.45uHy).

Gráfico 8

Measurement of the second coil (L = 7.58uHy).

3.2 Simulation

Again using the LTSpice software, the bandpass filter responses were simulated with the calculated values and with the components to be used in practice.

The first of these can be seen in graphs 9 and 10.

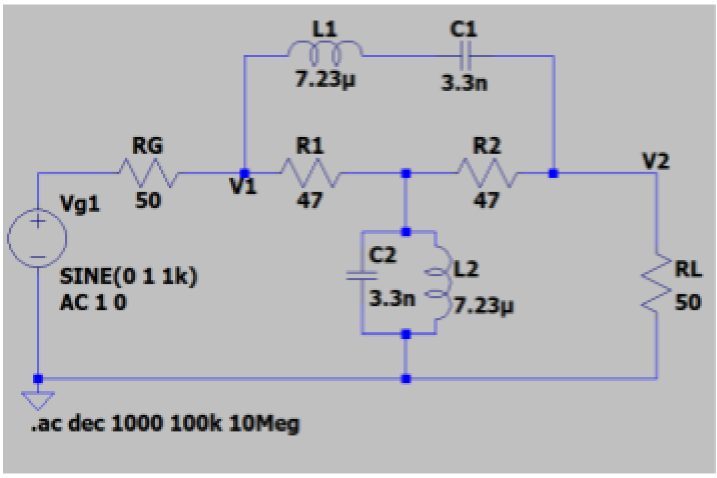


Gráfico 9

Calculated bandpass filter circuit.

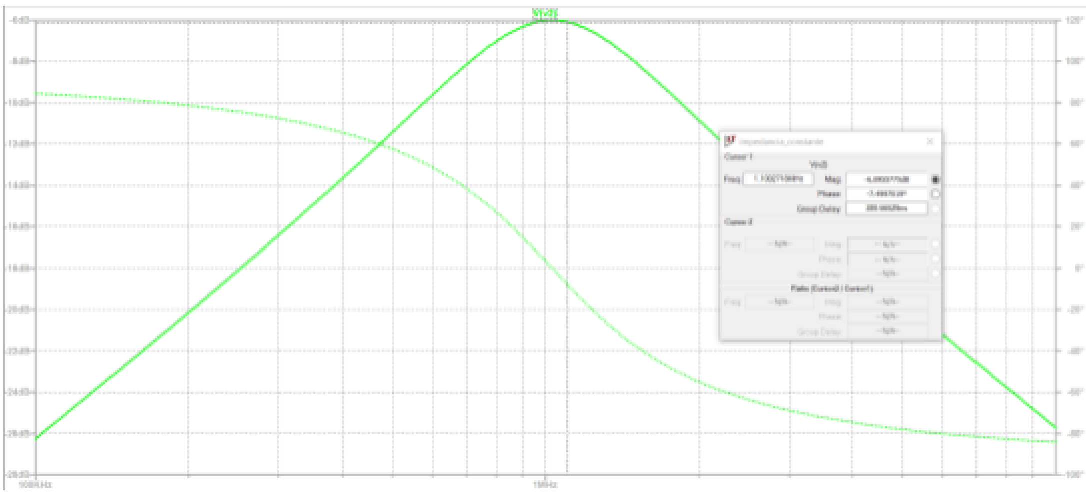


Gráfico 10

Frequency response of the filter calculated with att=-12.3734dB at fc.

In graphs 11 and 12 we present the version to be used in the laboratory.

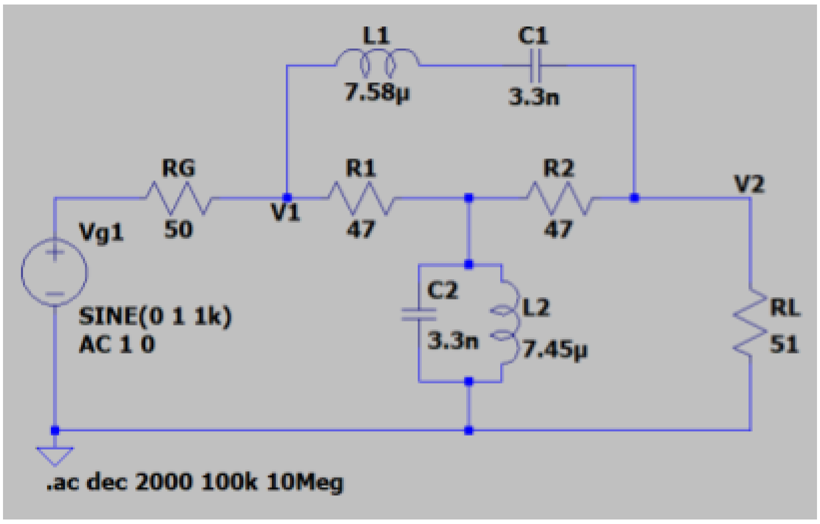


Gráfico 11

Bandpass filter circuit with commercial values.

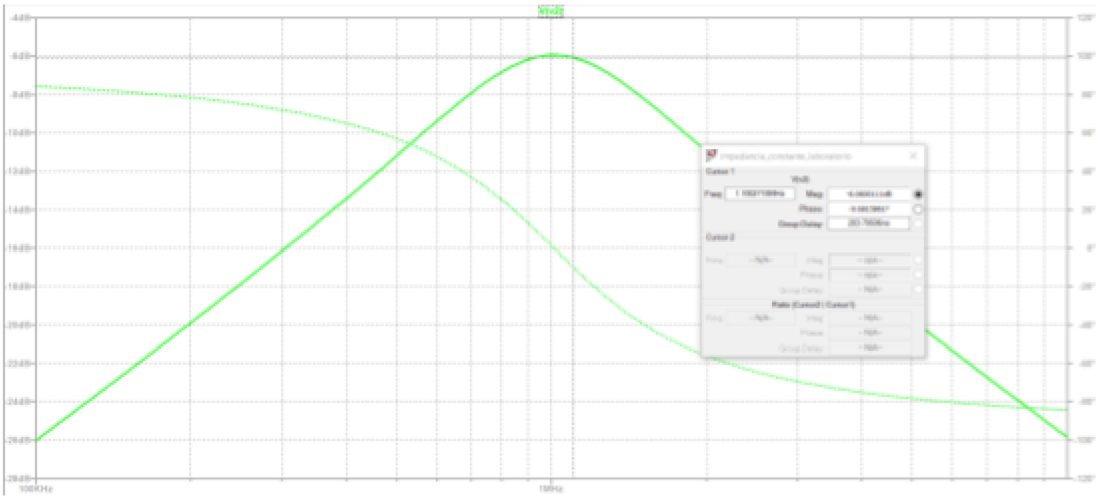


Gráfico 12

Frequency response of the commercial filter with att = -6.06dB at fc.

3.3 Measurements

<u>Part B</u>

The function generator was connected to the filter and its load, and 25 spaced points were taken to visualize the behavior at frequencies below the fc behavior in the fc and behavior above the fc.

The values of all points can be viewed in Table 3.

IN Frequency [kHz]	***IN amplitude [mVpp]***	***Amplitud OUT [mVpp]***	***Attenuation [dB]***
100	464	47	-19.89
200	472	94	-14.02
350	472	176	-8.57

Tabla 3

400	473	192	-7.83
500	472	240	-5.87
650	470	308	-3.67
700	474	340	-2.89
800	472	376	-1.98
950	472	416	-1.1
1000	456	420	-0.71
1100	456	428	-0.55
1250	458	400	-1.18
1300	456	404	-1.05
1400	454	388	-1.36
1550	456	364	-1.96
1600	456	340	-2.55
1700	455	322	-3
1850	440	288	-3.68
1900	424	272	-3.86
2000	424	258	-4.31
2150	424	232	-5.24
2200	426	220	-5.74
2300	400	212	-5.51
2450	400	192	-6.38
2500	405	192	-6.48
2800	360	158	-7.15
3000	360	144	-7.96
3400	350	122	-9.15
3800	300	100	-9.54
4000	300	96	-9.9

4400	276	80	-10.76
4800	276	71	-11.79
5000	260	70	-11.4

In graph 13 we present the comparison of the measurements made in the laboratory with what was expected in the simulation.

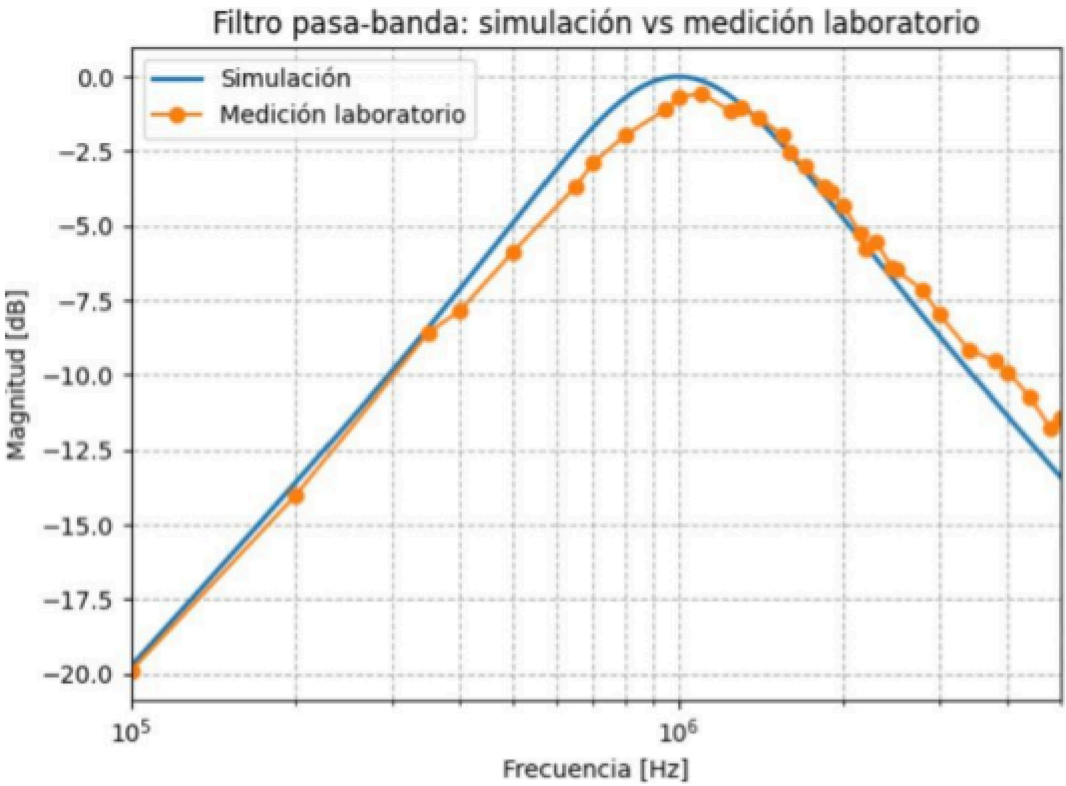


Gráfico 13

Comparison between simulated and measured values in the laboratory.

Next, images 3, 4 and 5 of measurements taken from the oscilloscope are attached.

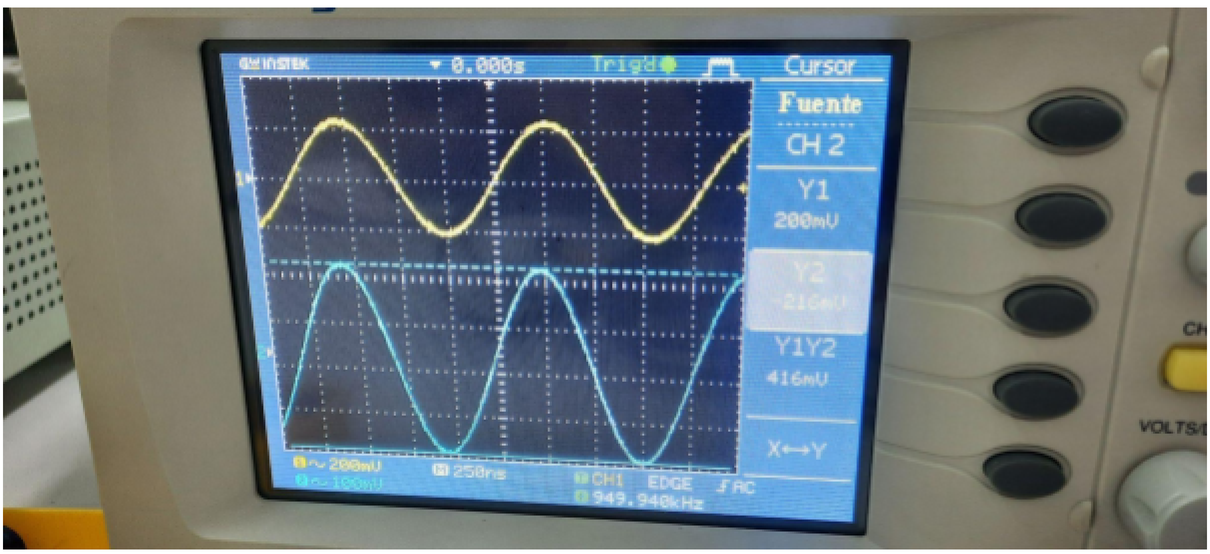


Imagen 3

Oscilloscope measurement of the voltage applied to the load at 950kHz.

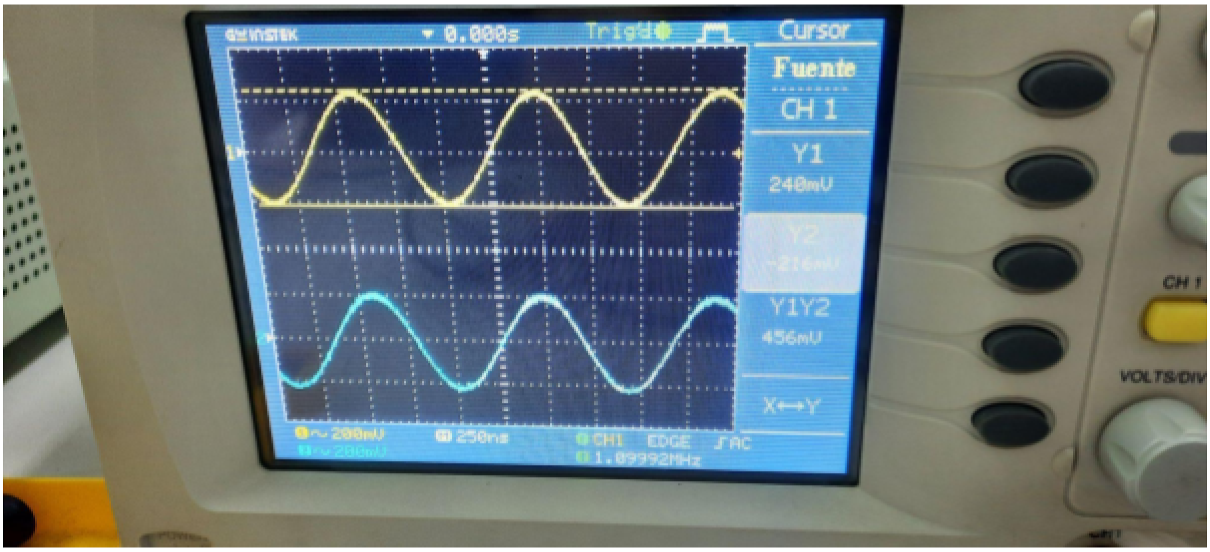


Imagen 4

Measurement on the oscilloscope of the voltage delivered by the generator at 1100kHz.

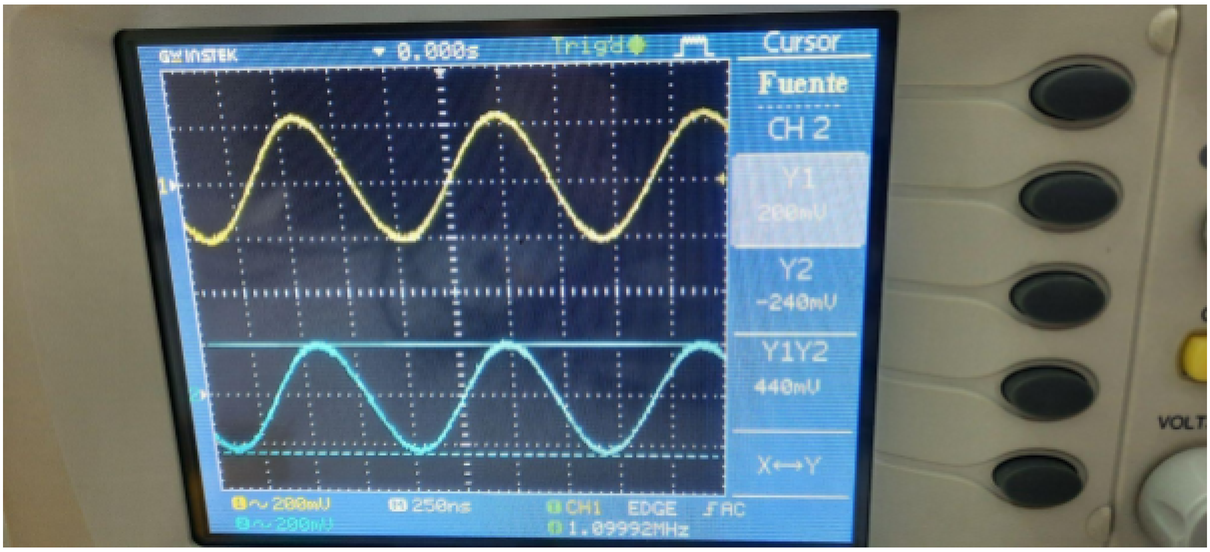


Imagen 5

Oscilloscope measurement of the voltage applied to the load at 1100kHz.

Part C

The filter was loaded with an impedance of about$510\Omega$and about 10 measurements were taken around the band. Then the process was repeated but with a$Z0 = 5\Omega$.

Table 4 presents the measurements for an impedance approximately 10 times greater, while Table 5 presents the measurements for an impedance 10 times smaller.

IN Frequency [kHz]	***IN Amplitude [mVpp]***	***Amplitud OUT [mVpp]***
10	488	12.3
100	480	82
500	456	424

Tabla 4

800	624	680
1000	820	792
1500	576	664
1800	456	536
2000	416	464
3000	360	256
4000	308	160
5000	288	112

IN Frequency [kHz]	***IN Amplitude [mVpp]***	***Amplitud OUT [mVpp]***
10	480	2.5
100	480	10
500	512	46
800	392	69
1000	146	74
1500	424	64
1800	480	55
2000	488	49
3000	432	27
4000	360	18
5000	312	12.8

Tabla 5

Continuing, in Table 6 these values were used to obtain the final response in dB for an impedance approximately 10 times greater. Meanwhile, in Table 7 these values were used to obtain the final response in dB for an impedance 10 times smaller.

IN Frequency [kHz]	***Attenuation [dB]***
10	-31.97
100	-15.35
500	-0.63
800	0.74
1000	-0.30
1500	1.23
1800	1.40
2000	0.95
3000	-2.96
4000	-5.69
5000	-8.20

Tabla 6

IN Frequency [kHz]	***Attenuation [dB]***
10	-45.66
100	-33.62
500	-20.93
800	-15.09
1000	-5.90
1500	-16.42
1800	-18.81
2000	-19.96
3000	-24.08
4000	-26.02
5000	-27.73

In graphs 14 and 15 we present the comparison of the measurements made in the laboratory with what was expected in the simulation.

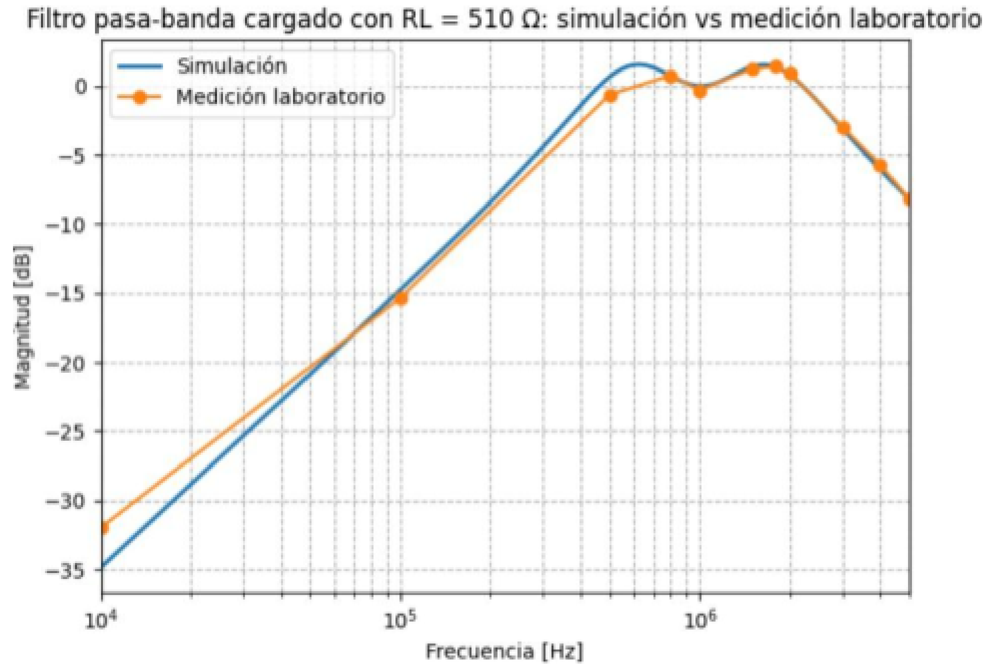


Gráfico 14

Comparison between simulated and measured values in the laboratory.

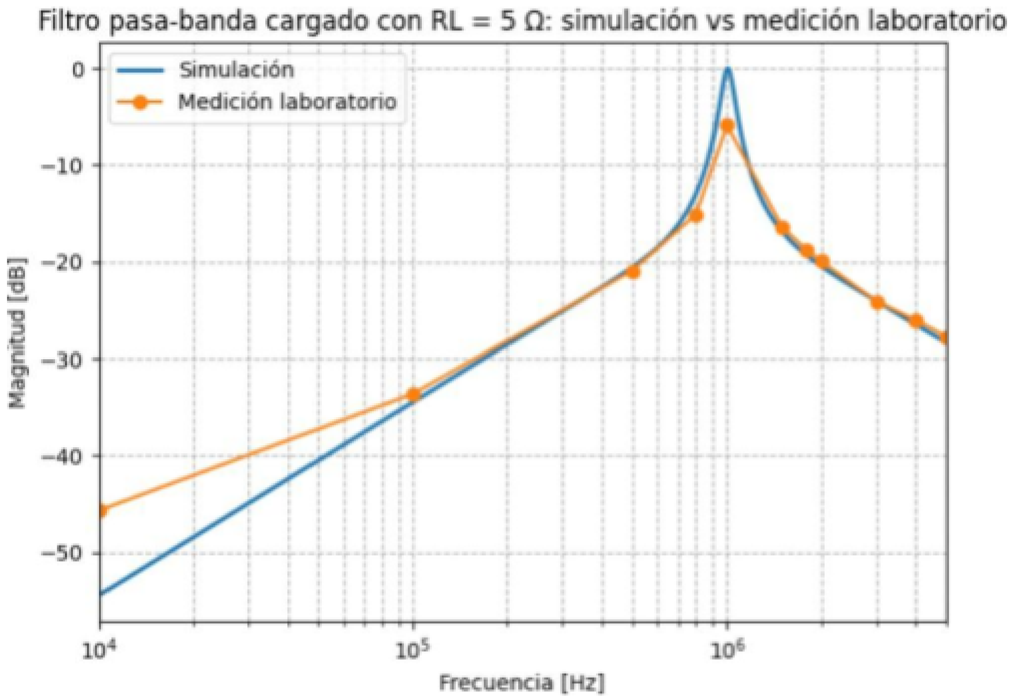


Gráfico 15

Comparison between simulated and measured values in the laboratory.

4. FULL FUNCTIONALITY

Finally, both circuits were used simultaneously to verify the operation of the project.

4.1 Measurements

Part D
The designed attenuator was added between the generator and the filter, and it was loaded with the$Z0 = 51\Omega$and the frequency characteristic was verified; Table 8 contains the measurements.

Finally, the terminal load was removed and the measurements of part C were repeated. Table 9 presents the results for a resistance 10 times larger and Table 10 for a resistance 10 times smaller.

IN Frequency [kHz]	*IN amplitude [mVpp]*	*Amplitud OUT [mVpp]*
10	464	2.8
100	472	3.8
500	464	16
800	458	25
1000	456	27.6
1500	440	23.6
1800	434	19
2000	432	16
3000	376	9.2
4000	340	9
5000	308	3.6

IN Frequency [kHz]	*IN Amplitude [mVpp]*	*Amplitud OUT [mVpp]*
10	480	2.8
100	484	7.2
500	472	28
800	472	43.2

1000	473	46.4
1500	440	41.5
1800	432	33.5
2000	424	28.3
3000	368	14.3
4000	320	8.4
5000	288	7.5

Tabla 9

IN Frequency [kHz]	***IN amplitude [mVpp]***	***Amplitud OUT [mVpp]***
10	472	4.3
100	472	5
500	472	4.5
800	456	9
1000	458	10
1500	424	6
1800	408	4
2000	400	3.44
3000	344	2.3
4000	278	3
5000	246	3.44

Next, in Table 11, these values were used to obtain the final dB response for the designed attenuator added between the generator and the filter, where the filter was loaded with the$Z0 = 51\Omega$Table 12 uses these values to obtain the final dB response for an impedance 10 times greater. Table 13 uses them for a resistance 10 times smaller.

Tabla 10

IN Frequency [kHz]	***Attenuation [dB]***
10	-44.39
100	-41.88
500	-29.25
800	-25.26
1000	-24.36
1500	-25.41
1800	-27.17
2000	-28.62
3000	-32.22
4000	-31.54
5000	-38.64

Tabla 11

IN Frequency [kHz]	***Attenuation [dB]***
10	-44.68
100	-36.55
500	-24.53
800	-20.77
1000	-20.16
1500	-20.51
1800	-22.21
2000	-23.51
3000	-28.21
4000	-31.62
5000	-31.68

IN Frequency [kHz]	Attenuation [dB]
10	-40.81
100	-39.50
500	-40.41
800	-34.09
1000	-33.22
1500	-36.98
1800	-40.17
2000	-41.31
3000	-43.49
4000	-39.34
5000	-37.08

In graphs 16, 17 and 18 we present the comparison of the measurements made in the laboratory with what was expected in the simulation.

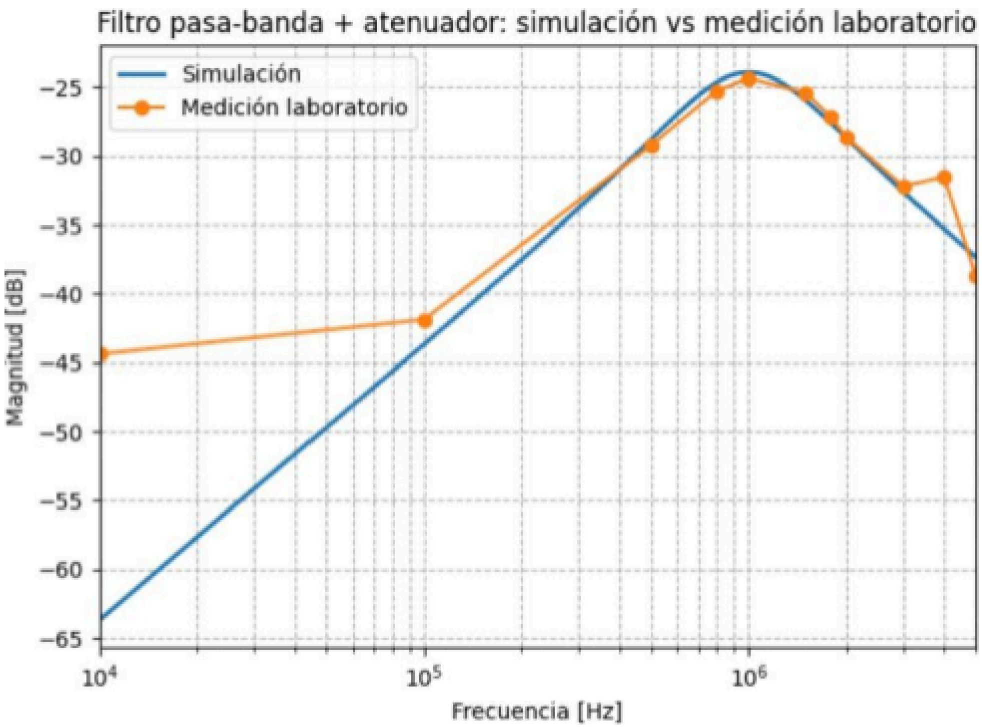


Gráfico 16

Comparison between simulated and measured values in the laboratory

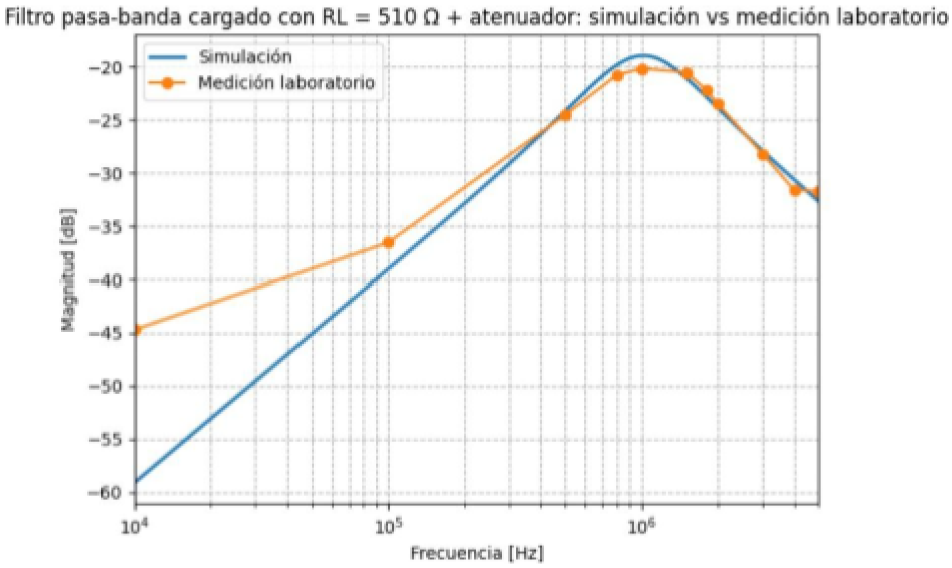


Gráfico 17

Comparison between simulated and measured values in the laboratory

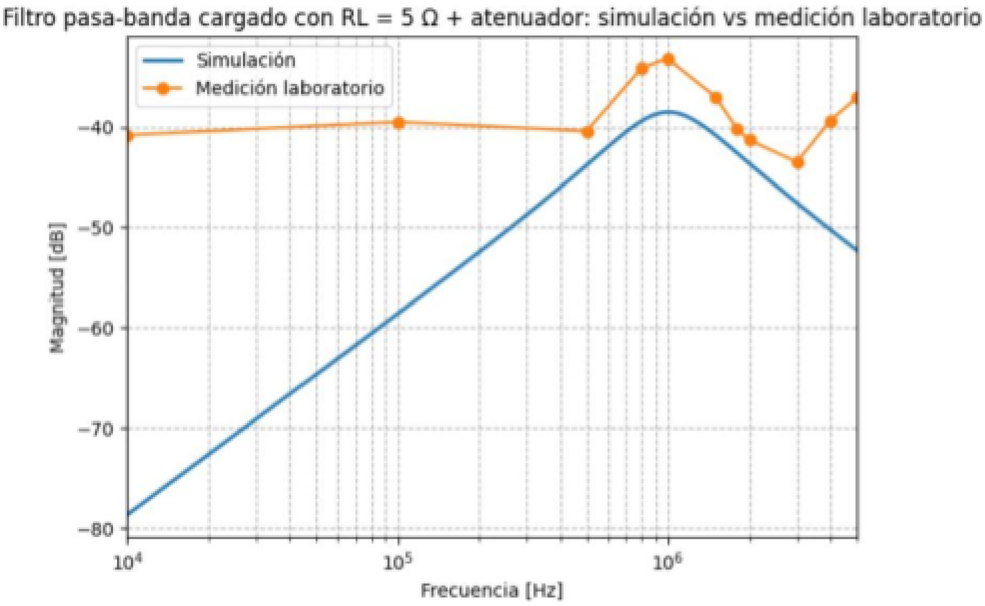


Gráfico 18

Comparison between simulated and measured values in the laboratory

5. CONCLUSIONS

5.1 Dissipative attenuator behavior

The dissipative attenuator and the constant impedance filter were successfully designed and implemented. In the case of the attenuator, the use of commercially available resistor values resulted in a slight discrepancy from the theoretical target of -24 dB, with measurements around -23.3 dB. This deviation is expected and acceptable considering the component tolerances and the adaptation to standard values. Regarding the filter, the manual fabrication of the inductors and their verification with the LCR bridge allowed the center frequency to be adjusted very close to the required 1100 kHz, validating the design method.

5.2 Behavior in response to load mismatch

When analyzing the filter's behavior under mismatch conditions (5 ohm and 510 ohm loads), we observed different phenomena:

- <u>**With a high impedance load of 510 ohms**</u>**:** The measured response closely resembles the simulation, maintaining the shape of the curve (Graphs 14 and 17).
- <u>**With a low impedance load of 5 ohms**</u>**:** The greatest discrepancy is observed here (Figures 15 and 18). While the ideal simulation predicts an attenuation of -80 dB, laboratory measurements "flattened" around -40 dB. This could be due to instrument noise and environmental interference that prevent the measurement of such small voltage values.

5.3 Cascade system analysis and measurement limitations

When connecting the quadripoles in cascade (Attenuator + Filter), it is observed that the phenomena seen when unbalancing the circuit individually are maintained and amplified in the joint connection.

As shown in Figure 18, i.e., in cascade with a 5-ohm load, the simulation and measurement diverge significantly in the rejection bands. This is due to two main factors:

1. **Signal level and instrumental error:**With a -24 dB input attenuator, the signal reaching the filter is already on the order of tens of millivolts. When attenuated again by the mismatched filter, the resulting voltages drop into the microvolt range, where the oscilloscope's measurement error is much higher and the signal becomes indistinguishable from background noise.
2. **Parasites of the montage:**At high frequencies and such low voltages, the parasitic capacitances and inductances of the components during measurement (circuit board, cables, solder joints, distance between components, etc.) indicate alterations in the ideal behavior of the components, creating alternative signal paths that prevent achieving the infinite attenuation predicted by theory. In other words, there is a leakage effect.

In summary, the design meets the transfer and bandwidth specifications, but it reveals the physical limitations of the laboratory implementation when working with very low amplitude signals and high rejection impedances.